

New Dispo GoLive 1060 (Oracle)

Date: 2026-04-17

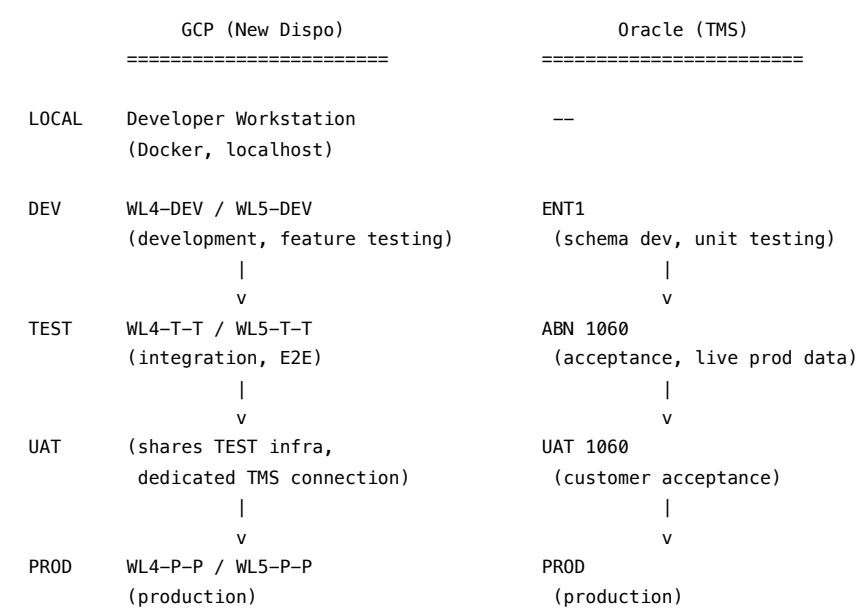
Status: Draft

Purpose: Holistic overview to align all stakeholders on the architecture, infrastructure, environments, and ownership for Branch 1060 on Oracle.

1. End-to-End Environment Landscape

This section maps every environment stage across both the GCP (New Dispo) side and the Oracle (TMS) side.

1.1 Environment Pipeline Overview



1.2 Environment Mapping Matrix

Stage	GCP Project (WL4)	GCP Project (WL5)	Oracle Instance	Data Profile	Sign-Off
LOCAL	--	--	--	Seeded / empty	Developer
DEV	WL4-DEV (TBD)	WL5-DEV (TBD)	ENT1 (shared)	Schema only, no branch data	Developer
TEST	prj-cal-w-wl4-t-4c48-53ad	prj-cal-w-wl5-t-6c00-53ad	ORA-ABN-1060	Live production data (1060)	Patrick U., Max K. (P3)
UAT	(shares TEST infra)	(shares TEST infra)	ORA-UAT-1060	Production data	Max Beisheim, Patrick U.
PROD	prj-cal-w-wl4-p-afad-53ad	prj-cal-w-wl5-p-3e5b-53ad	PROD	Production	--

2. Architecture Overview

2.1 GCP Workload Separation

Workload	Purpose	Components	GCP Project (Test)	GCP Project (Prod)
WL4	User-Facing App	Frontend, Backend, CloudSQL	prj-cal-w-wl4-t-4c48-53ad	prj-cal-w-wl4-p-afad-53ad
WL5	Integration & Data	TMS Bridge, Cloud4Log, Dispo Filter, CrossDock	prj-cal-w-wl5-t-6c00-53ad	prj-cal-w-wl5-p-3e5b-53ad
Network	Shared VPC	VPN, Routing, Firewall	prj-cal-net-s-t-e004-53ad	prj-cal-net-s-p-19c3-53ad
CI/CD	Deployment	Workload Identity, Pipelines	prj-cal-w-cicd-wl4-a1bc-53ad / prj-cal-w-cicd-wl5-a591-53ad	(same)

3. GCP Components by Environment

3.1 Cloud Run Services

Service	Workload	Test Name	Test URL	Prod Name
Frontend	WL4	cal-new-disposition-frontend-t-t	https://test.dispo.gcp.nagel-group.com	cal-new-disposition-fron
Backend	WL4	cal-new-disposition-backend-t-t	https://test.dispo.gcp.nagel-group.com	cal-new-disposition-back
TMS Bridge	WL5	cal-new-disposition-tmsbridge-t-t	https://test.tms-bridge.gcp.nagel-group.com	cal-new-disposition-tmsb

3.2 Cloud Functions (Gen2)

Function	Workload	Trigger	Test Instance	Prod Instance	Config
Dispo Filter (UAT1060)	WL5	Cloud Storage	new-dispo-filter-shipment-records-uat1060	new-dispo-filter-shipment-records-uat1060	Python

3.3 Databases

Database	Type	Test Instance	Prod Instance
New Dispo Backend DB	CloudSQL PostgreSQL	cal-new-disposition-psql-t-t	cal-new-disposition-postgres-p-p

3.6 Cloud Storage Buckets

Bucket	Environment	Purpose	Status
WL5_CDC_BUCKET_ABN1060	Test	CDC bucket for ABN1060	TBD
WL5_CDC_BUCKET_UAT1060	Test	CDC bucket for UAT1060	TBD
WL5_CDC_BUCKET_1060	Prod	CDC bucket for 1060	TBD

3.7 Pub/Sub

Resource	Environment	Purpose	Status
WL5_CDC_TOPIC_ABN1060	Test	CDC events for ABN1060	TBD

Resource	Environment	Purpose	Status
WL5_CDC_TOPIC_UAT1060	Test	CDC events for UAT1060	TBD
WL5_CDC_TOPIC_1060	Prod	CDC events for 1060	TBD

3.8 Networking

Resource	Test	Prod	Status
Shared VPC	vpc-c-shared-vpc-c-net-s-t	vpc-c-shared-vpc-c-net-s-p	* to be confirmed
Subnet	sn-vpc-c-net-s-t-europe-west3-common	sn-vpc-c-net-s-p-europe-west3-common	* to be confirmed
Hub VPC Project	prj-cal-net-h-5332-53ad	prj-cal-net-h-5332-53ad	* to be confirmed
CAL VPN Endpoints	34.157.54.59 , 34.157.191.34	(same hub)	* to be confirmed
Nagel VPN Endpoints	35.242.18.84 , 34.157.189.239	(same hub)	* to be confirmed

4. Security - Users & Service Accounts

4.1 Authentication Patterns by Context

Context	Method	Identity Provider	Status
User login (Browser)	OAuth2 Authorization Code + PKCE	Keycloak	* to be verified
Service-to-Service (Cloud)	Client Credentials (OAuth2)	Keycloak	* to be verified
CI/CD Pipeline	Workload Identity Federation	GCP IAM (Azure DevOps token exchange)	* to be verified
Cloud Run <-> CloudSQL	CloudSQL Proxy + IAM	GCP IAM	* to be verified
Cloud Functions <-> Resources	GCP Service Account	GCP IAM	* to be verified
Local Development	Username/Password	Local Keycloak + local Postgres	* to be verified

4.2 Keycloak Clients

Client ID	Type	Used By	Environment	Status
cal-client	User-facing (Auth Code)	Frontend + Backend	Production	* to be verified
ebv-client	User-facing	Legacy EBV integration	Production	* to be verified
client-credentials-test	Machine-to-Machine	Service accounts	Test	* to be verified
cloud-run-client	Machine-to-Machine	Cloud Run functions	Cloud environments	* to be verified
tms-cloud-service	Machine-to-Machine	TMS Bridge	Cloud environments	* to be verified

Keycloak Servers:

Environment	URL
Local	http://localhost:8080
Dev	https://dev.new-dispo.nagel.p3ds.net/keycloak
Production	https://nagel-staging.ddns.net:8081/keycloak

4.3 GCP Service Accounts

Service Account	Scope	Purpose	Status
wl-cicd@prj-cal-w-cicd-wl4-a1bc-53ad.iam.gserviceaccount.com	WL4 CI/CD	Azure DevOps pipeline deployments (Frontend, Backend)	* to be confirmed
wl-cicd@prj-cal-w-cicd-wl5-a591-53ad.iam.gserviceaccount.com	WL5 CI/CD	Azure DevOps pipeline deployments (TMS Bridge, Cloud4Log, CDC)	* to be confirmed
wl5-cloudrun@prj-cal-w-wl5-t-6c00-53ad.iam.gserviceaccount.com	WL5 Test Runtime	Cloud Run + Workflow execution (Test)	* to be confirmed
wl5-cloudrun@prj-cal-w-wl5-p-3e5b-53ad.iam.gserviceaccount.com	WL5 Prod Runtime	Cloud Run + Workflow execution (Prod)	* to be confirmed

Workload Identity Pools (Azure DevOps Federation):

- WL4: azure-devops (project number: 607166292072)
- WL5: azure-devops (project number: 534233136072)

4.4 Database Users

Database	User (expected)	Context	Notes
ORA-ABN-1060	TMS1060 / TMSBR1060	TMS Bridge -> ORA-ABN-1060	Pending: connection details from Joachim
ORA-UAT-1060	TMS1060 / TMSBR1060	TMS Bridge -> ORA-UAT-1060	Pending: after ABN sign-off
ORA-PROD-1060	TMS1060 / TMSBR1060	TMS Bridge -> ORA-PROD-1060	Pending: after UAT sign-off

Database Identifier Convention (ADR-004):

Format: {DBMS}-{COUNTRY}-{COMPANY}-{BRANCH} (e.g., 0-D-10-60 for Oracle Germany Company 10 Branch 60)

* to be discussed in scope

4.5 Secret Manager

Secret Name	Purpose	Injected Into
D-{COMPANY}-{BRANCH}	PostgreSQL connection string	TMS Bridge
O-{COMPANY}-{BRANCH}	Oracle connection string	TMS Bridge

5. External Integrations

System	Protocol	Purpose	Connected From	Owner
Keycloak	HTTPS (OIDC)	Authentication & Authorization	Frontend, Backend, TMS Bridge	CAL / Nagel
Azure Service Bus	AMQP/TLS	Event publishing for CALSuite	CrossDock Publisher	CAL
TOP Service	HTTP	Route optimization	Backend (on-prem)	Nagel
xServer	HTTP	Routing calculations	TOP Service	PTV
SMTP (Office 365)	STARTTLS	Email notifications	Backend	CAL
Timocom	REST API	Freight exchange	Backend	External
Trans.eu	REST API (OAuth2)	Freight exchange	Backend	External

6. Ownership & Responsibility Matrix (WIP)

6.1 Go-Live 1060 -- Who Does What

#	Task	Owner	Support	Status	Notes
1	Oracle ENT1 schema development	Joachim Schreiner (Nagel)	--	Active	Wrapper procedures for 1060
2	Provision ORA-ABN-1060	Bernd Friedewald, Thomas Paulus (Nagel)	Joachim	In Progress	Live production data from 1060
3	Provision ORA-UAT-1060	Bernd Friedewald, Thomas Paulus (Nagel)	Joachim	Pending	After ABN sign-off
4	Oracle deployment pipeline (QS tool)	Joachim Schreiner (Nagel)	--	Operational	ENT -> ABN -> UAT -> PROD
5	ORA-ABN-1060 connection details	Joachim Schreiner (Nagel)	--	Pending	Host, port, credentials, network
6	TMS Bridge config for ORA-ABN-1060	P3 (Matthias, Max K.)	Joachim	Pending	Blocked on #5
7	Backend TEST env config for Oracle	P3 (Matthias, Max K.)	--	Pending	Blocked on #6
8	GCP Secret Manager: Oracle connection strings	P3 / CAL Infra	--	Pending	0--{COMPANY}--{BRANCH} entries
9	Network/VPN: Oracle on-prem reachable from GCP	CAL Infra / Nagel Infra	--	TBD	Verify oracle-user tag grants access to 1060
10	End-to-end integration test (ABN 1060)	P3 (Matthias, Max K.)	Joachim	Pending	Frontend -> Backend -> Bridge -> ORA-ABN-1060
11	Character encoding validation (UTF-8 vs Oracle)	P3	Joachim	Pending	Polish/special-char data in ABN 1060
12	TMS Pulse load test (ABN 1060)	P3	Nagel	Pending	Requires ABN with real data
13	ABN sign-off	Patrick U., Max K. (P3)	--	Pending	Gate to UAT
14	UAT sign-off	Max Beisheim, Patrick U. (Nagel)	--	Pending	Gate to PROD
15	Oracle CDC pipeline (Striim/Datastream)	TBD	--	Open Question	Should CDC connect to ABN/UAT too?
16	Dispo Filter function for 1060 CDC	P3	--	Pending	New function instance per branch
17	Pub/Sub topic for 1060 CDC	P3 / CAL Infra	--	Pending	WL5_CDC_TOPIC_1060
18	Pipeline testing to Production WL4	P3	--	Open Question	Verify CI/CD pipelines deploy to Prod WL4

6.2 Standing Ownership

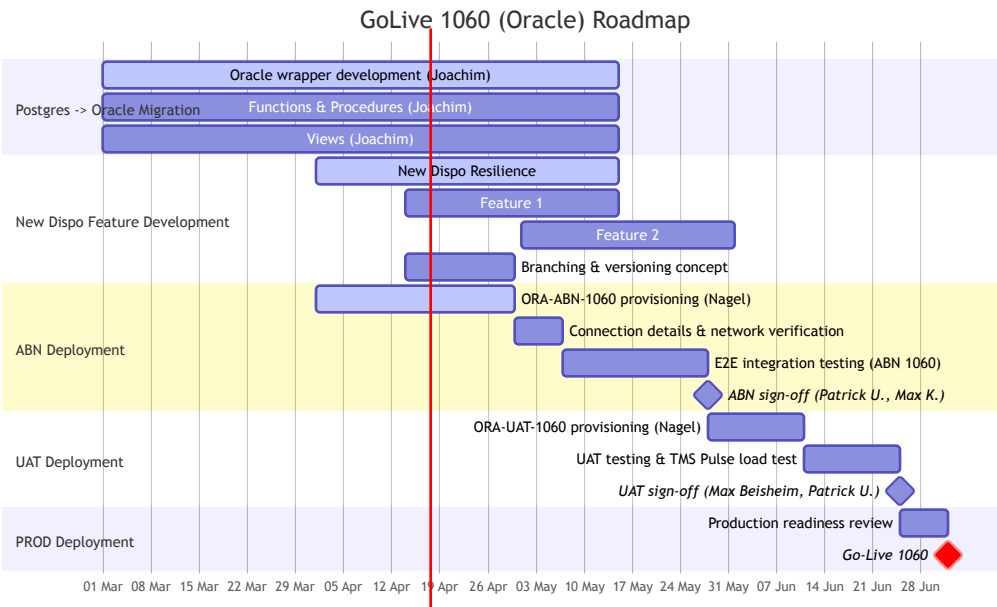
Area	Owner
GCP Infrastructure (WL3/WL4/WL5)	P3 (?)
New Dispo Frontend	P3
New Dispo Backend	P3
TMS Bridge	P3
TMS AlloyDB Schema	P3
TMS Oracle Schema / Wrappers	Nagel (end-to-end)

Area	Owner
Oracle Dev & Deployment Pipeline	Nagel (end-to-end)
Oracle Instance Provisioning	Nagel (end-to-end)
Keycloak	P3
VPN / Network	Nagel Platform
Azure DevOps Pipelines	P3
Freight Exchanges (Timocom, Trans.eu)	P3

7. Roadmap*

To be created by Patrick U. and Max K.

All dates are indicative and subject to alignment with Patrick U. (PO) and Max K. (TechPO).
 Database deployments follow a **2-week release cycle** (abn1034, uat1034, and 1060 DBs going forward).



8. CI/CD & Release Branches

Branching & versioning concept currently in the making.

9. Risks & Open Items (WIP)

#	Risk / Open Item	Impact	Mitigation	Owner
1	ORA-ABN-1060 availability date unknown	Blocks all integration testing	Coordinate with Joachim for timeline	Matthias
2	ORA-ABN-1060 connection details pending	Blocks TMS Bridge config	Obtain from Joachim once provisioned	Matthias

#	Risk / Open Item	Impact	Mitigation	Owner
3	Character encoding (UTF-8 vs Oracle legacy)	Data corruption (Poland incident)	Explicit testing with Polish data in ABN 1060	P3
4	Wrapper edge cases on real 1060 data	Runtime failures in production	ABN 1060 has real data -- test early	P3 + Joachim
5	VPN/Network path to Oracle 1060 from GCP	TMS Bridge cannot reach Oracle	Verify with CAL Infra	CAL Infra
6	Oracle CDC pipeline for 1060 not scoped	CDC events missing for 1060 branches	Decide: Striim vs Datastream for Oracle CDC	TBD
7	Sign-off criteria undefined	Unclear go/no-go gate	Define criteria for ABN and UAT	Patrick U.
8	Packet loss GCP <-> Nagel on-prem	Intermittent failures	Managed by Telekom/Arista -- monitor	CAL Infra

10. Next Steps (WIP)

1. **Obtain ORA-ABN-1060 connection details** from Joachim (host, port, user, network path)
2. **Verify VPN/network connectivity** from GCP WL5 to Oracle 1060 on-prem
3. **Create GCP secrets** for Oracle 1060 connection (0-D-10-60 pattern)
4. **Configure TMS Bridge** appsettings for ORA-ABN-1060
5. **Run first end-to-end integration test** through full stack
6. **Test character encoding** with Polish/special-character data
7. **Scope Oracle CDC** for 1060 (Striim/Datastream decision)
8. **TMS Pulse load test** against ABN 1060
9. **ABN sign-off** -> proceed to UAT 1060
10. **Clarify application access methods** per environment (Citrix for UAT, Cameyo for ABN, direct access in Nuremberg)
11. **UAT sign-off** -> go-live

11. Related Resources

Resource	Location
Oracle Environment Assessment	02_Explorations/2026-04-10_Oracle_DEV_ABN_UAT_Instances_for_Branch_1060/
Infrastructure Documentation	08_Documentation/Infrastructure/
GCP Resources Reference	08_Documentation/Infrastructure/gcp-resources.md
Deployment Mapping	08_Documentation/Infrastructure/deployment-mapping.md
Network Configuration	08_Documentation/Infrastructure/network-configuration.md
Technical Platform Overview	02_Explorations/2026-04-08_Technische_Plattform_Ubersicht/
GCP Workloads Diagram	07_Diagrams/GCP-workloads.svg
Wiki: Environments	WIKI/Nagel-CAL-Disposition.wiki/Devops/Environments.md
ADR-004: DB Identifier Convention	01_ADRs/